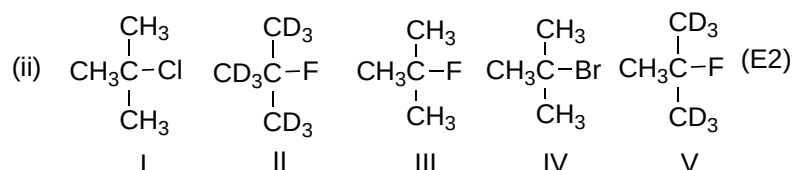
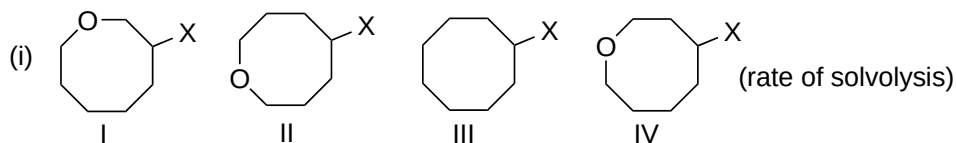


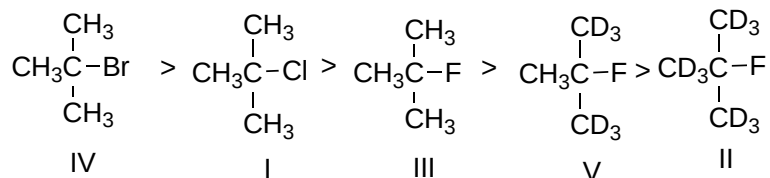
ACML100 Tutorial 4
Substitution and Elimination reactions

Q1. Arrange the following compounds in increasing order of their reactivity.

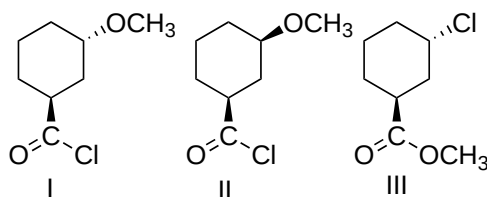


Ans. (i) I < IV < III < II

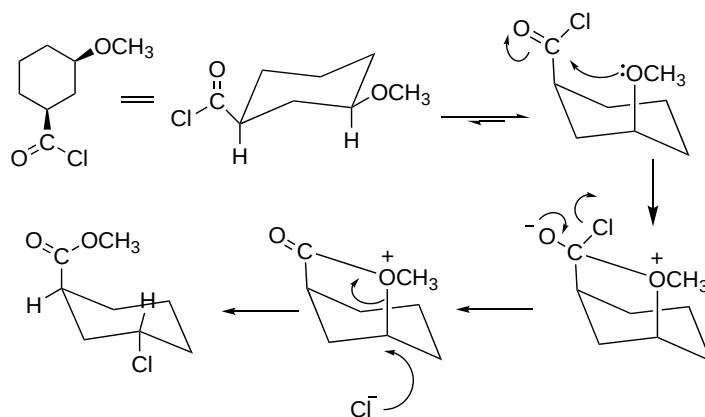
(ii) Br⁻ is good leaving group and F⁻ (a stronger base), is a bad leaving group. It is easier to break C-H bond than a C-D bond. So more the hydrogens on β-carbons are replaced by deuterium atom, slower the rate of E2 will be.



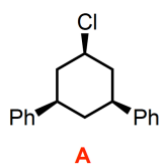
Q2. Which one of the compounds (**I** or **II**) reacts with Cl⁻ to give (**III**)? Give the mechanism of its formation.



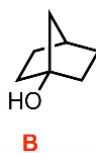
Ans. The compound **II** will give the **III** via NGP.



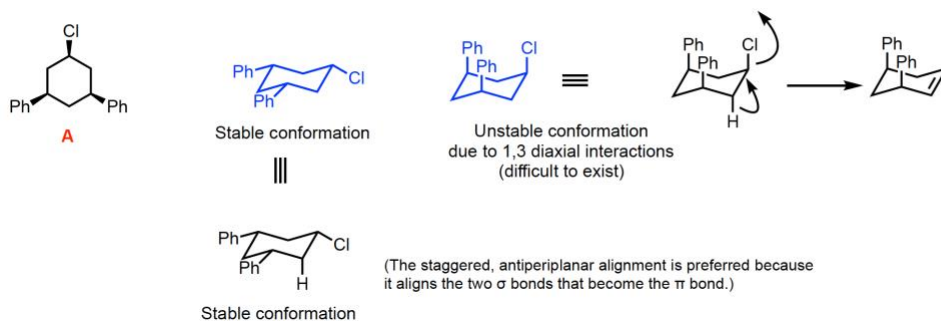
Q3. a) Explain why the compound A does not undergo an E2 elimination with strong base.



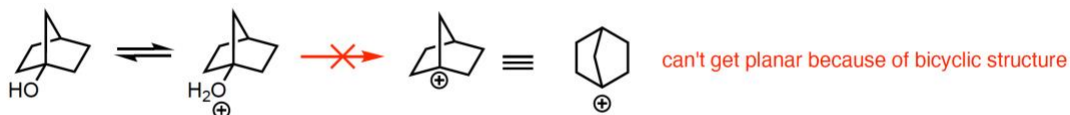
b) Explain why the compound B do not undergo an E1 elimination with strong acid.



Ans.



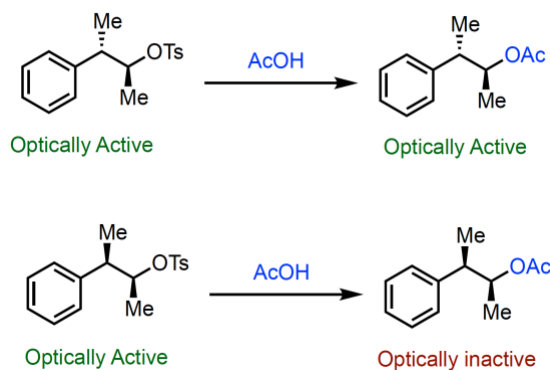
E₂ reactions occur most rapidly when the H-C bond and C-LG bonds involved are co-planar, most often at 180° with respect to each other. This is described as an antiperiplanar conformation. This conformation positions the σ bonds that are being broken in the correct alignment to become the π bond.



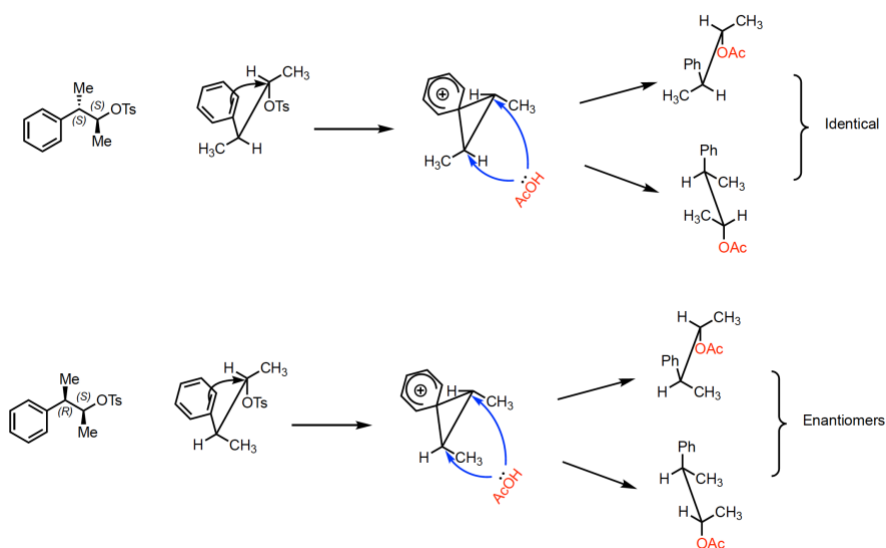
The bicyclic structure prevents the bridgehead carbon becoming planar so, although the cation would be tertiary, it is very high in energy and does not form.

Bredt's rule: double bonds can never be formed to bridgehead carbons in bicyclic systems.

Q4. Explain the following observation with the suitable mechanism.



Ans.



Q5. Arrange the order of reactivity of the following compounds to the action of aq. alkali and suggest reasons: chlorobenzene, vinyl chloride, allyl chloride.

Q6. Arrange the following series of ions in increasing order of stability.

(i) $(\text{CH}_3)_3\text{C}^+$, $(\text{CH}_3)_2\text{CH}^+$, CH_3CH_2^+ .

(ii) $(\text{C}_6\text{H}_5)_3\text{C}^+$, $(\text{C}_6\text{H}_5)_2\text{CH}^+$, $\text{C}_6\text{H}_5\text{CH}_2^+$.

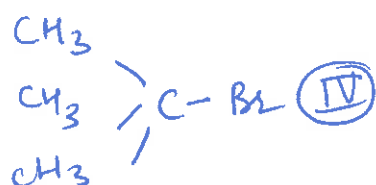
(iii) $p\text{-CH}_3\text{OC}_6\text{H}_4\text{CH}_2^+$, $p\text{-ClC}_6\text{H}_4\text{CH}_2^+$, $p\text{-O}_2\text{NC}_6\text{H}_4\text{CH}_2^+$.

Ans 5.

allyl chloride > vinyl chloride > chlorobenzene



Ans 6.



I > II > III > IV

Reaction goes via SN²
NaOEt is a strong
base.